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CAMBRIDGE
INTERNATIONAL EXAMINATIONS

June 2003

INTERNATIONAL GCSE

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 0625/06

PHYSICS
Alternative to Practical

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1	**(a)**		Seven correct values: 0, 2, 3, 6, 9, 10, 12 (-1 each error)		
	(b)		Graph:		
			Scales, labelled, suitable size		1
			Axes, right way round		1
			Plots to ½ sq (-1 each error)		2
	(c)		Line shape		1
			Line thickness		1
			Triangle greater than ½ line and method used		1
			Correct interpolation to ½ sq		1
				TOTAL	**10**
2	**(a)**		36° ($\pm 1^o$)		1
	(b)		Refracted ray drawn		1
			22° ($\pm 1^o$)		1
			normal correct (by eye)		1
			neat, thin, correct lines		1
	(c)		Correct refracted ray (by eye) with arrow		1
	(d)		Separation (LHS) at least 5cm		1
			Separation (RHS) at least 5cm		1
				TOTAL	**8**
3	**(a)**	**(i)**	Voltmeter across lamp		1
		(ii)	Variable resistor/rheostat		1
	(b)		Correct position		1
	(c)		V		1
			A		1
			Ω		1
			correct R at 9.8V = 8.16666 (any sf)		1
			all R to 2/3 sf		1
			consistent 2 sf or consistent 3 sf		1
				TOTAL	**9**
4	**(a)**	**(i)**	6.8cm (68mm)		1
		(ii)	6.8		1
			unit, mm		1
	(b)	**(i)**	3.8/3.77 or 0.38/0.377		1
			mm or cm as appropriate		1

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		(ii)	0.94/0.95 (or evidence of division by 4)	1
		(iii)	0.75094/0.75095	1
	(c)		Thickness of string/thickness of marks on string/stretching of string/metre rule measures to 1mm	1
			TOTAL	**8**
5	**(a)**	**(i)**	polystyrene	1
		(ii)	Least steep curve (or numbers suitably quoted)	1
	(b)		Three from: Thickness of insulator Room temp. Starting temp. Mass/vol./amount of water Using same can	3
			TOTAL	**5**

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Grade thresholds taken for Syllabus 0625 (Physics) in the June 2003 examination.

	maximum mark available	minimum mark required for grade:			
		A	C	E	F
Component 1	40	-	29	23	19
Component 2	80	-	45	34	26
Component 3	80	53	32	-	-
Component 5	60	42	33	21	13
Component 6	40	34	26	20	15

The threshold (minimum mark) for B is set halfway between those for Grades A and C.
The threshold (minimum mark) for D is set halfway between those for Grades C and E.
The threshold (minimum mark) for G is set as many marks below the F threshold as the E threshold is above it.
Grade A* does not exist at the level of an individual component.